Yiren Xu

PhD Candidate in Statistics, University of California, Irvine
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Education

University of California, Irvine **PhD in Statistics** Expected 2026
Advisor: Daniel L. Gillen
Dissertation: "Statistical methods for maximizing information for subgroup effects estimation"
University of California, Irvine **M.S. in Statistics** Dec. 2023
University of California, Los Angeles **B.S. in Statistics**, Minor in English Jun. 2021

Research Interests

Survival analysis, clinical trial design, efficient sampling designs, group sequential designs, Alzheimer's disease research, recruitment science

Research Experience

Intern, Clinical Biometrics, Cytokinetics Inc. Jun. – Sep. 2024
Simulations using Bayesian dynamic borrowing to leverage historical data in a clinical trial

- Utilized innovative Bayesian borrowing methods with time-to-event outcomes through simulations in R, JAGS, and rstan to plan for future clinical trial
- Simulated power increase while controlling type I error of an adaptive design with interim analyses by leveraging historical trial data
- Presented results and visualizations in team- and company-wide meetings with 200+ colleagues

Graduate Student Researcher, UCI Statistics Jul. 2022 – Present
Sampling designs in survival analysis to improve efficiency for detecting subgroup differences

- Modified the case-cohort (CC) design under the Cox model to increase efficiency to detect subgroup differences, relevant in Alzheimer's disease (AD) biomarker discovery
- Implemented simulations in R to show increased power in assessing subgroup effects, especially in rare subpopulations, while conserving biomarker samples by up to 70%

Estimation framework, sample size and power calculation for recurrent event CC designs

- Conduct simulations in R to compare sampling strategies and propose method that increases power in later events while maintaining efficiency in marginal model
- Derive theoretical formulas for power and sample size to optimize study planning
- Present results quarterly to faculty advisor and 10+ statistics students

Researcher, UCI Institute for Memory Impairments and Neurological Disorders Jul. 2022 – Present
Research attitudes and retention in a recruitment registry

- Performed data cleaning and analysis (logistic, survival models) in R on longitudinal registry data
- Collaborated with 5+ statisticians, neuroscience researchers, and data manager in weekly meetings to draft and publish manuscript

Study partner requirement in Alzheimer's disease preclinical trials

- Conduct analysis in R on survey data (logistic, CART methods) and create visualizations using ggplot2 to identify and describe potential subgroup differences in recruitment barriers
- Lead project in cross-functional team and submit to peer-reviewed journals and AD conferences

Publications

Witbracht, M. G., **Xu, Y.**, Morgan, O. B., Salazar, C. R., Hoang, D., Kind, A., Gillen, D. L., & Grill, J. D. (2025). Research Attitudes Questionnaire scores and retention in a recruitment registry. *Journal of Alzheimer's Disease*, doi.org/10.1177/13872877241302422.

In preparation:

Xu Y., Grill J. D., & Gillen D. L. Case-cohort designs to improve efficiency for detecting effect modification.

Xu Y. & Gillen D. L. Increasing efficiency in case-cohort designs for recurrent events analysis.

Xu, Y., Witbracht, M. G., Glover, C. M., Gillen, D. L., & Grill, J. D. Attitudes toward the study partner requirement in Alzheimer's disease preclinical trials among diverse populations.

Presentations

Attitudes toward the study partner requirement in Alzheimer's disease preclinical trials among racially and ethnically diverse populations

- Poster, *Clinical Trials on Alzheimer's Disease (CTAD)*. San Diego, CA, Nov. 2025

Increasing efficiency in case-cohort designs for recurrent events analysis

- Contributed talk, *Joint Statistical Meetings (JSM)*. Nashville, TN, Aug. 2025

Novel study designs towards precision medicine in Alzheimer's Disease research

- Poster, *REMIND Emerging Scientists Symposium*. Irvine, CA, Feb. 2025

Case-cohort designs to improve efficiency for detecting effect modification

- Presentation, *WNAR of the International Biometric Society*. Fort Collins, CO, Jun. 2024

Representation in research: from recruitment to analysis

- Presentation, *UCI Steckler Center (CREATE) Briefings.* Irvine, CA, May 2024

Assessing renewal in a local recruitment registry

- Poster, *REMIND Emerging Scientists Symposium*. Irvine, CA, Apr. 2023

UCI Statistical Methods and Applied Research Talks (SMART) Seminar Series. Irvine, CA, 2022 – Present

Teaching

UCI Statistics, Basic Statistics

Teaching Assistant Jan. – Jun., Oct. – Dec. 2022

Reader Oct. – Dec. 2021

Awards & Service

UCI 2025-26 Public Impact Fellow Dec. 2025 – Present

UCI Steckler Center for Responsible, Ethical, and Accessible Technology (CREATE)

Fellow, Graduate Researcher Jan. 2023 – Present

UCI Diverse Educational Community and Doctoral Experience (DECADE)

Department Mentorship Program Mentor/Student Representative Oct. 2021 – Present

Statistical & Programming Languages

R, Python

Professional Memberships

Phi Beta Kappa Honor Society, American Statistical Association, International Biometric Society, Alzheimer's Association ISTAART